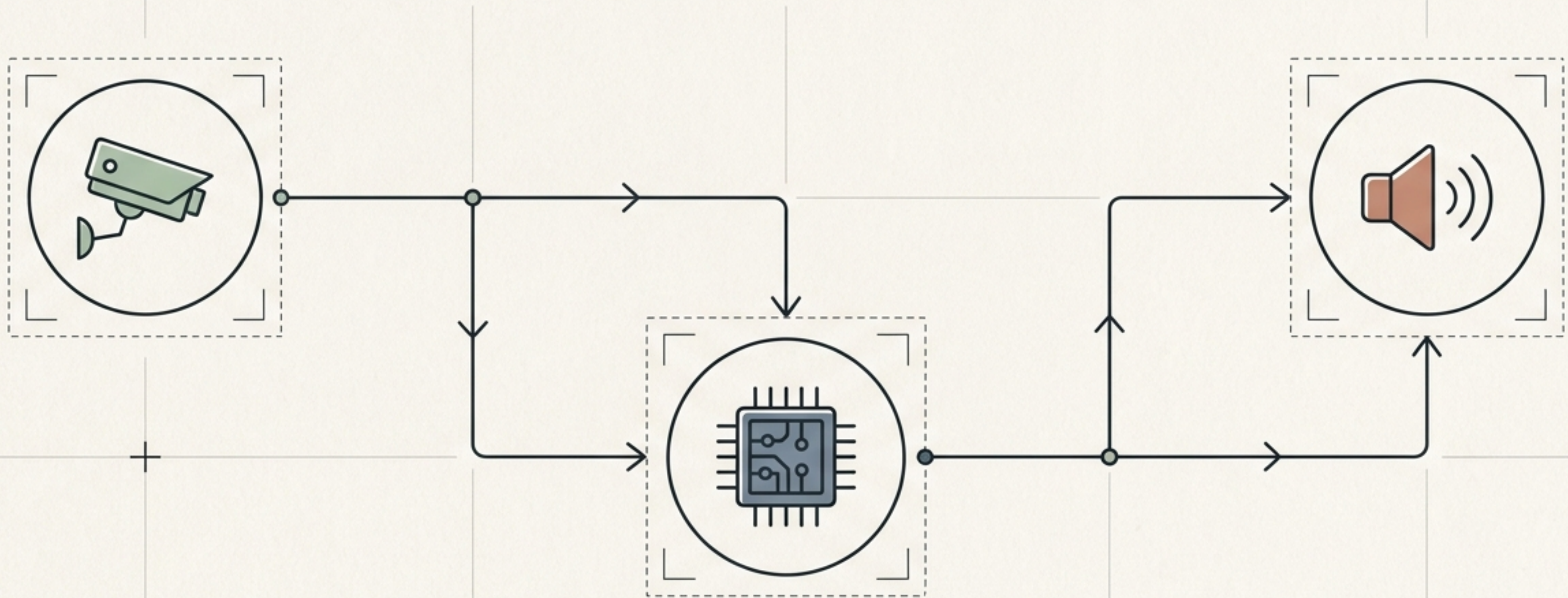


EdgeGuard AI: Event-Driven, Privacy-First Edge Security



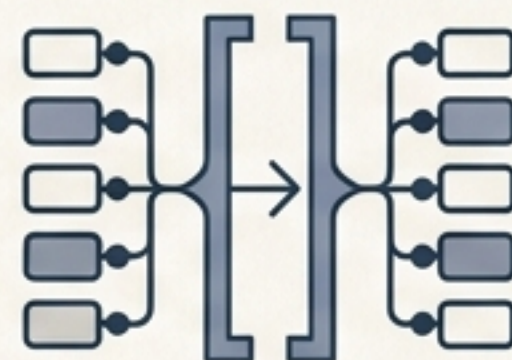
Turning Ordinary Sensors into One Vocal, Event-Driven System

Processing everything on the LAN without relying on cloud logic.



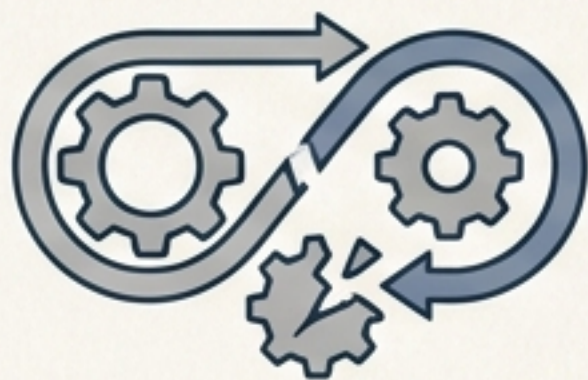
Local-First

Face detection/recognition runs on an NVIDIA Jetson Xavier NX. Video and embeddings never leave the LAN.



Event-Driven

No hardwired commands. Every observation becomes a structured JSON event on an MQTT bus.



Reliability

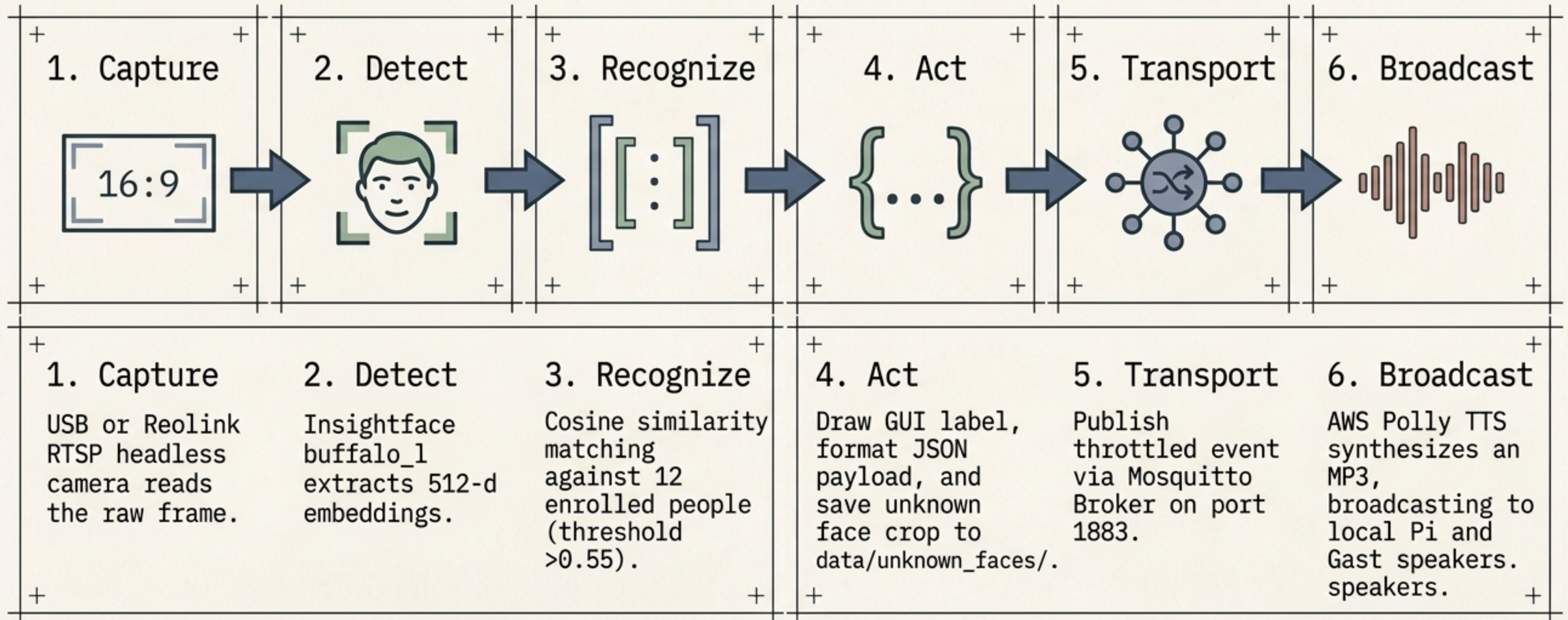
Never crash the pipeline. If a broker, speaker, or cloud service fails, the system logs, skips, and keeps watching.



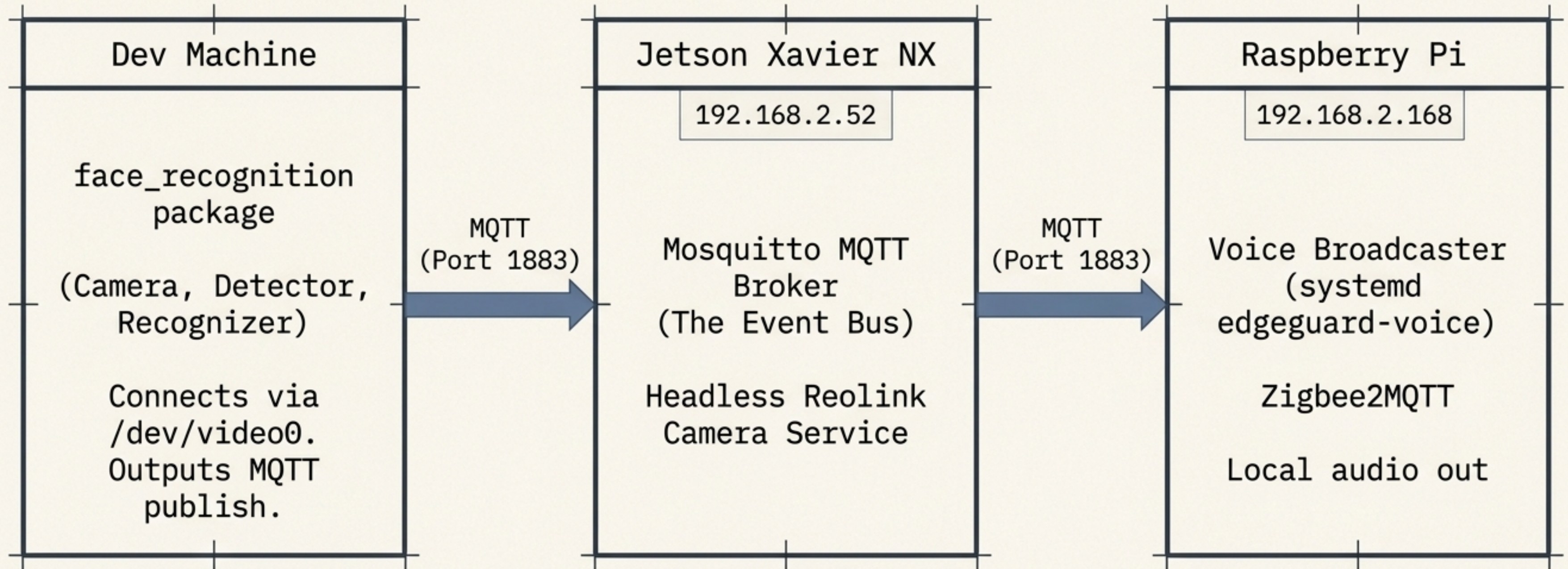
Privacy by Design

No face data, reference embeddings (faces.npz), or credentials ever hit Git or the cloud.

The Event Lifecycle: From Lens to Loudspeaker



System Architecture: The Modularity Map



KEY PRINCIPLE: No shared state. Coupling is achieved exclusively through MQTT topics and JSON schemas. Any machine can run any service.

Anatomy of an Event Payload

```
{  
  "event_id": "evt_28cae03f3fe8",  
  "event_type": "unknown_person_detected",  
  "source": "camera",  
  "camera_id": "front_entrance",  
  "confidence": 0.71  
}
```

Triggers the AWS Polly template logic to formulate a security alert. Known persons (person_recognized) are logged but silently suppressed by default.

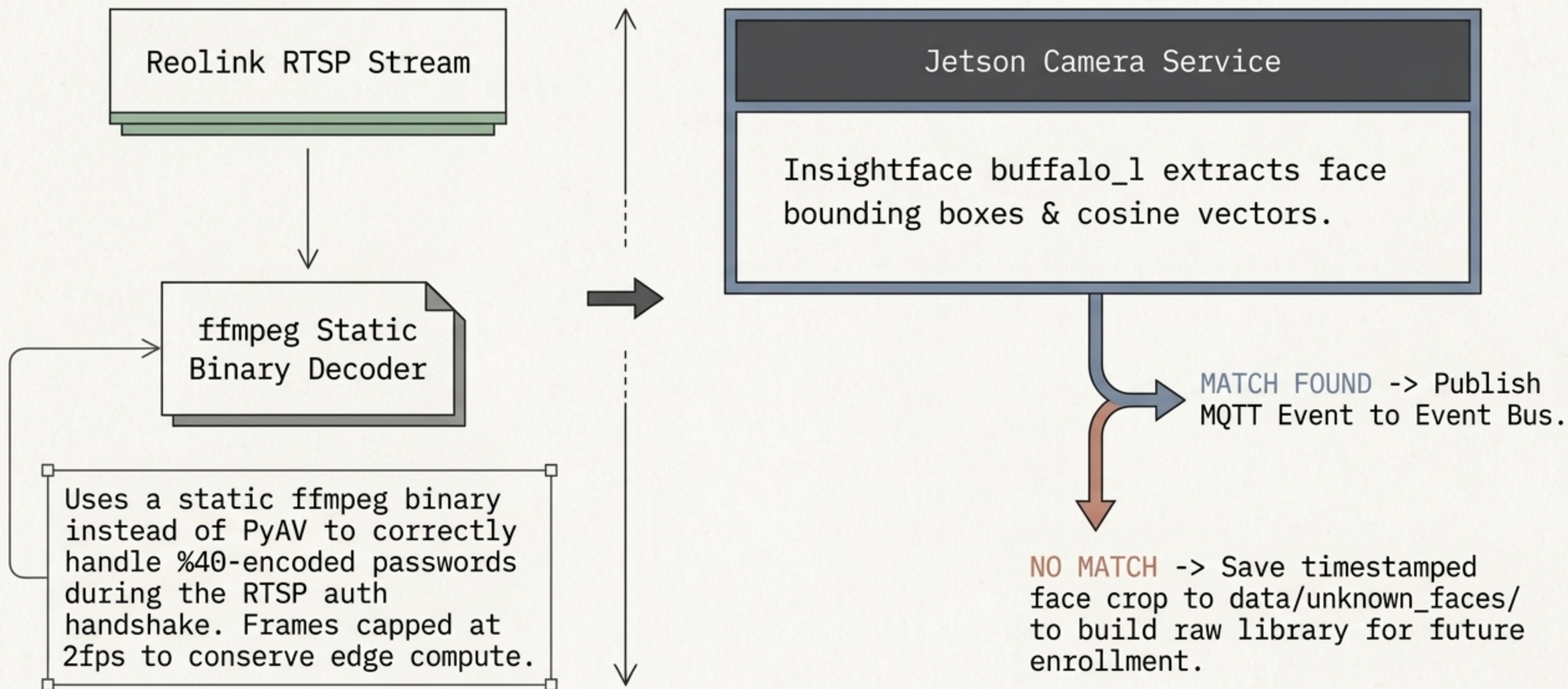
Dynamically maps to voice/config.py to inject human-readable locations into synthesized speech ('..at the front entrance').

Published to edgeguard/camera/<camera_id>. Throttled to re-publish only every 10 seconds while the person remains in frame.

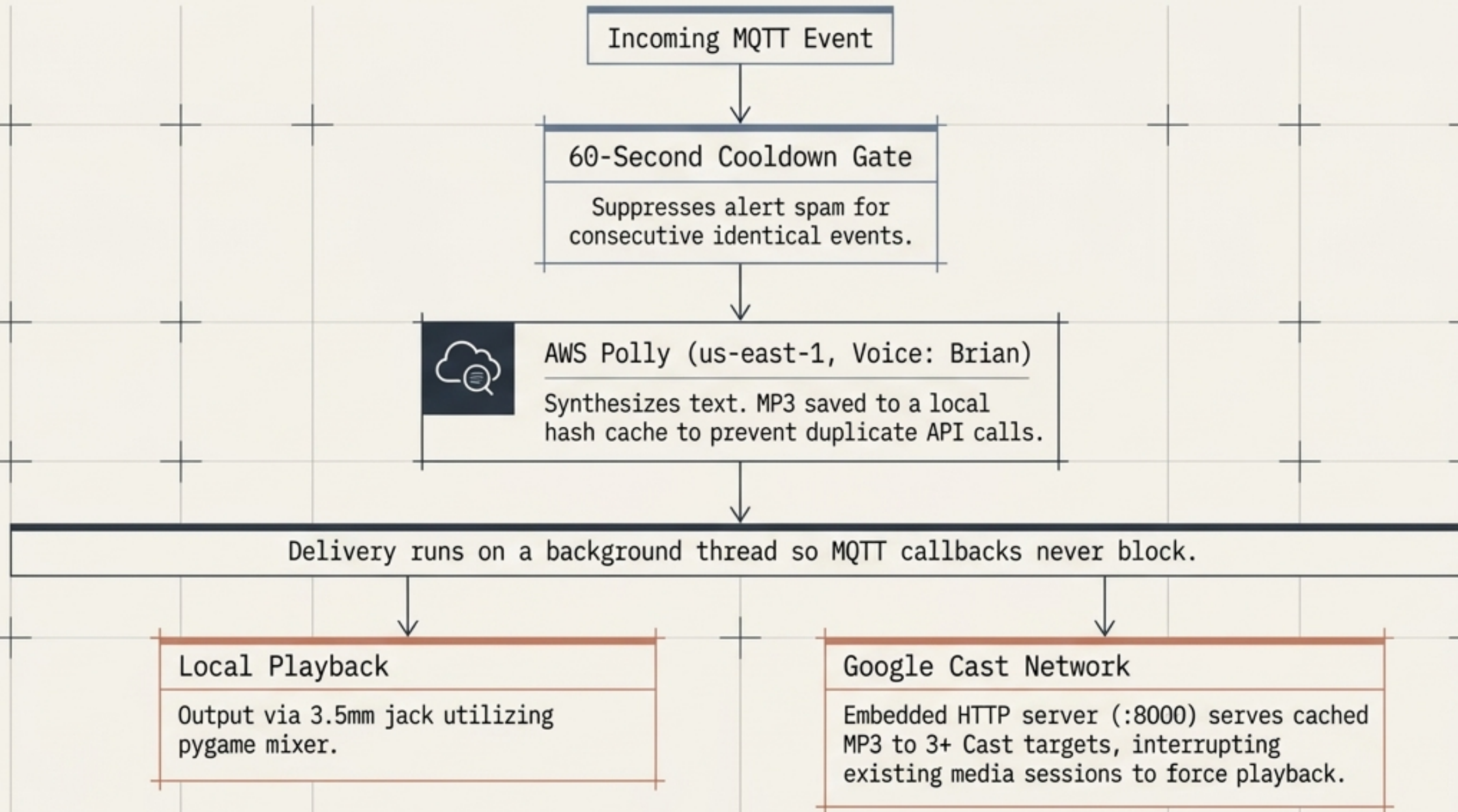
Deployment Topology & Hardware Matrix

Hardware Node & IP	System Role	Key Technical Stack
Jetson Xavier NX 192.168.2.52	Event Bus Hub & Vision Compute	CUDA 10.2, TensorRT 7.1, Ubuntu 18.04
Raspberry Pi 3 192.168.2.168	Voice Broadcaster & Zigbee Gateway	Raspbian 11, Docker, systemd, pygame
Reolink IP Cameras (x2) 192.168.2.183 / 184	Yard/Garage Observation Input	Headless RTSP (imageio-ffmpeg), 960x720 @ 2fps
Google Cast Speakers (x4)	Announcement Delivery	Discovered by exact friendly name (e.g., 'Kitchen speaker'); immune to DHCP IP changes.
Zigbee Sensors (SONOFF)	Door Contacts & Plugs	Bridged via Zigbee2MQTT, reporting battery and state changes.

The Headless Vision Pipeline



The Voice Delivery Engine



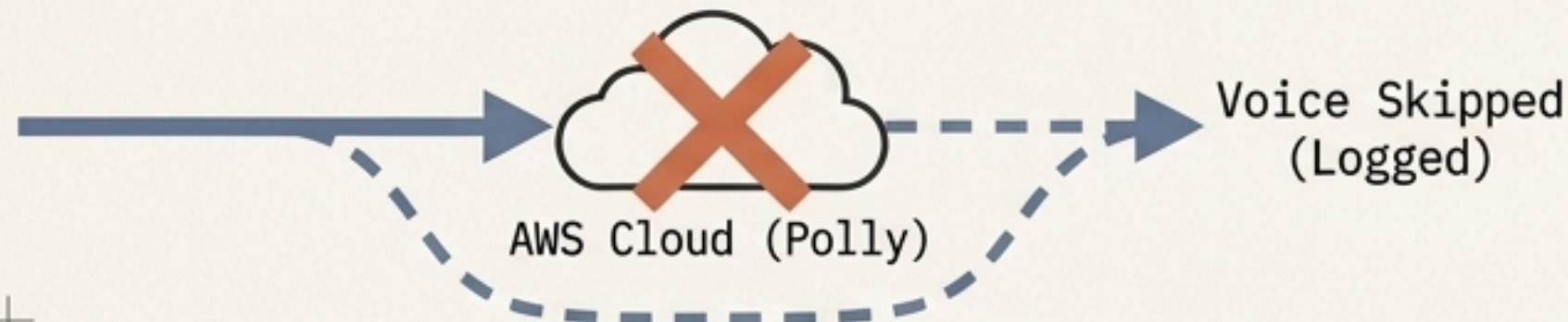
Reliability by Design: Failure Semantics

Scenario 1: MQTT Broker Down



Camera pipeline logs dropped events, caches state, and auto-reconnects with 1-30s backoff. Vision pipeline never crashes.

Scenario 2: AWS Polly Down / Bad Creds



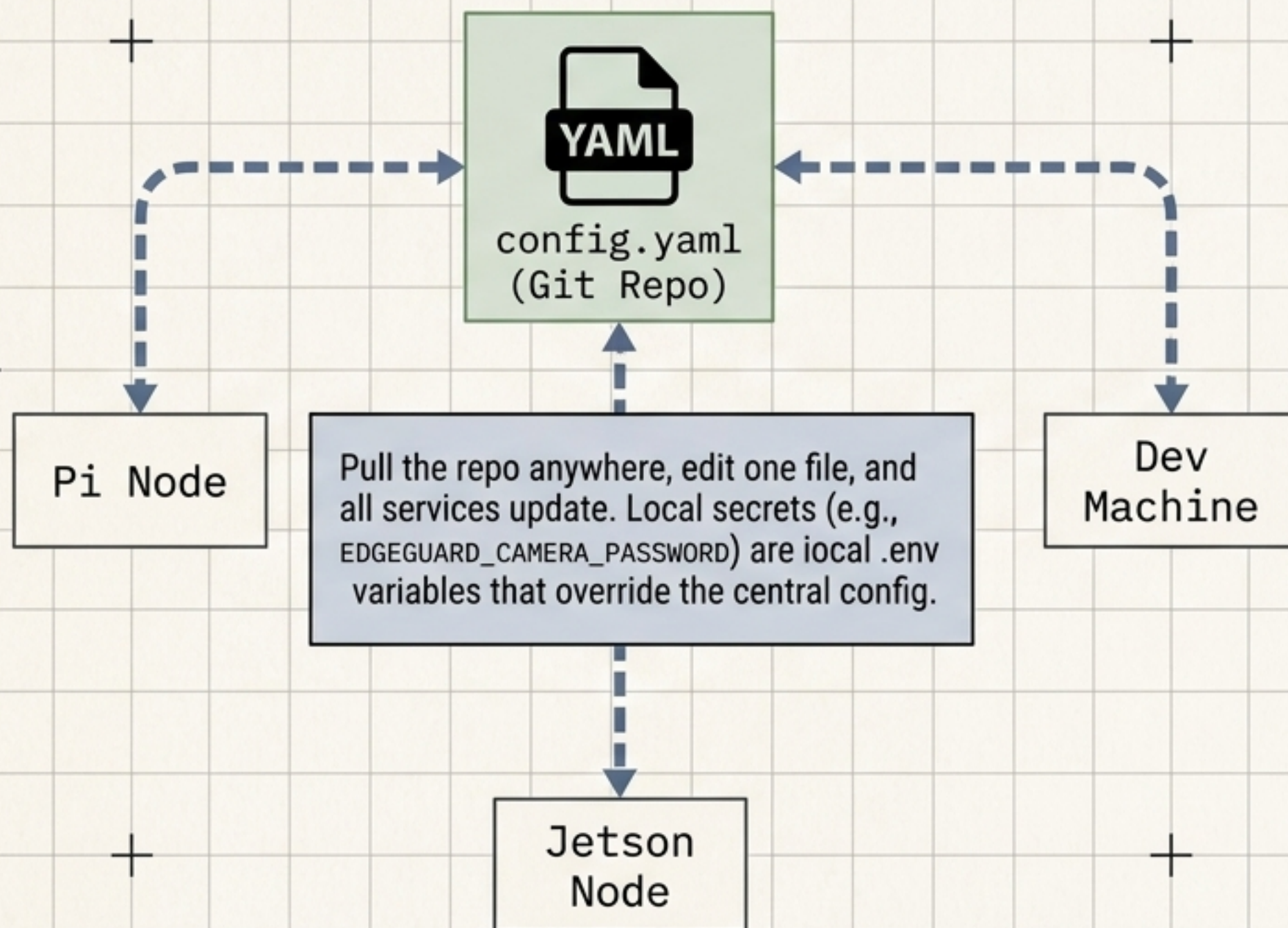
Voice announcement is gracefully skipped. The event continues to be logged. Camera observation is completely unaffected.

Scenario 3: A Cast Speaker Goes Offline



Total isolation. Remaining Cast speakers and the physical Pi speaker still play the announcement perfectly.

One Config, Every Device & Roadmap



Status & Roadmap Checklist

- ☒ Edge Vision & Multi-Camera integration deployed.
- ☒ Dual-Channel Voice Broadcast (Local + Cast) active.
- ☒ 15 legacy scheduled announcements ported to the new event bus.
- ☐ Zigbee sensor MQTT correlation (bridging door contacts to cameras).
- ☐ Unified Security Dashboard and Armed/Disarmed state engine.